

LUMINARY ACADEMY

Mathematics Workbook

Grade 12 Mathematics

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Chapter 1

Functions — Advanced Topics

Learning Objectives

- Define and evaluate composite and inverse functions
- Analyse exponential and logarithmic functions
- Apply the laws of logarithms
- Sketch and interpret transformation of functions

Key Formulas

- Composite: $(f \circ g)(x) = f(g(x))$
- Inverse: f^{-1} exists when f is one-to-one (injective)
- $ff^{-1}(x) = x$ and $f^{-1}f(x) = x$
- Exponential: $f(x) = a^x$, $a > 0$, $a \neq 1$
- Logarithm: $\log_a(x) = y \iff a^y = x$
- Laws of Logs: $\log_a(MN) = \log_a M + \log_a N$
- $\log_a(M/N) = \log_a M - \log_a N$
- $\log_a(M^n) = n \log_a M$
- Change of base: $\log_a b = \ln(b)/\ln(a)$
- Natural log: $\ln(e^x) = x$ and $e^{(\ln x)} = x$

1.1 Worked Examples

Example 1

Given $f(x) = 2x + 1$ and $g(x) = x^2$, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

Solution:

$$(f \circ g)(x) = f(x^2) = 2x^2 + 1 \quad (g \circ f)(x) = g(2x+1) = (2x+1)^2 = 4x^2 + 4x + 1$$

Example 2

Find the inverse of $f(x) = (3x - 2) / (x + 1)$.

Solution:

$$\text{Let } y = (3x-2)/(x+1) \quad y(x+1) = 3x-2 \quad xy + y = 3x-2 \quad xy - 3x = -2-y \quad x(y-3) = -2-y \quad x = (-2-y)/(y-3) = (2+y)/(3-y) \quad f^{-1}(x) = (x+2)/(3-x)$$

Example 3

Simplify: $\log(27) - \log(3) + \log(9)$

Solution:

$$\log(27) = 3, \log(3) = 1, \log(9) = 2 \quad 3 - 1 + 2 = 4$$

Example 4

Solve: $2^{x+1} = 32$

Solution:

$$2^{x+1} = 2^{x+1} \quad x+1 = 5 \quad x = 4$$

Example 5

Solve: $\log(x) + \log(x-2) = 3$

Solution:

$$\log(x(x-2)) = 3 \quad x(x-2) = 8 \quad x^2 - 2x - 8 = 0 \quad (x-4)(x+2) = 0 \quad x = 4 \text{ (reject } x = -2, \text{ outside domain)}$$

Common Mistakes

- $\log(A + B) \neq \log(A) + \log(B)$ — the log laws apply to products, not sums!
- Always check domain when solving log equations — argument must be positive.

1.2 Practice

Independent Practice

1. $f(x) = x + 3$, $g(x) = 2x$. Find $(f \circ g)(4)$ and $(g \circ f)(4)$.

Answer: _____

2. Find f^{-1} for: $f(x) = 5x - 7$

Answer: _____

3. Evaluate: $\log_{10}(1000) - \log_{10}(10)$

Answer: _____

4. Solve: $3^x = 81$

Answer: _____

5. Simplify: $\log_2(8) + \log_2(4) - \log_2(2)$

Answer: _____

6. Solve: $\log(x + 5) + \log(x - 2) = 1$ (base 10)

Answer: _____

7. Write in logarithmic form: $5^3 = 125$

Answer: _____

8. Sketch $y = 2^x$ and $y = \log_2(x)$ on the same axes — what do you notice?

Answer: _____

9. Solve: $e^{2x} = 7$

Answer: _____

10. If $f(x) = 3x$ and $g(x) = x - 2$, find $(f \circ g)^{-1}(x)$.

Answer: _____

1.3 Exit Ticket

Exit Ticket

1. Find $(f \circ g)(2)$ if $f(x) = x^2$ and $g(x) = x + 1$.

Answer: _____

2. Write in exponential form: $\log_2(64) = 3$

Answer: _____

3. Solve: $2^x = 16$

Answer: _____

4. What is $\ln(e^2)$?

Answer: _____

5. Is $f(x) = x^2$ one-to-one? Why?

Answer: _____

Chapter 2

Sequences, Series & Financial Mathematics

Learning Objectives

- Identify and use arithmetic and geometric sequences
- Find sum of arithmetic and geometric series (finite and infinite)
- Solve compound interest and annuity problems
- Apply sequences to real-world financial contexts

Key Formulas

- Arithmetic: $T_n = a + (n-1)d$ | $S_n = n/2 \times (2a + (n-1)d)$ or $n/2 \times (a + l)$
- Geometric: $T_n = ar^{n-1}$ | $S_n = a(r^n - 1)/(r - 1)$ for $r \neq 1$
- Infinite Geometric: $S_\infty = a/(1-r)$, valid when $|r| < 1$
- Compound Interest: $A = P(1 + r/n)^{nt}$
- Continuous Compounding: $A = Pe^{rt}$
- Present Value: $PV = FV/(1 + r)^n$

2.1 Worked Examples

Example 1

Arithmetic sequence: first term 3, common difference 5. Find T_{15} and S_{15} .

Solution:

$$T_{15} = 3 + 7(5) = 3 + 35 = 38 \quad S_{15} = 15/2 \times (2(3) + 14(5)) = 7.5 \times (6 + 70) = 7.5 \times 76 = 570$$

Example 2

Geometric sequence: 2, 6, 18, 54, ... Find T_n and S_n .

Solution:

$$a = 2, r = 3 \quad T_n = 2 \times 3^{n-1} = 2 \times 3^{12} = 1458 \quad S_n = 2(3^n - 1)/(3 - 1) = 2(2187 - 1)/2 = 2186$$

Example 3

Find sum of infinite geometric series: $12 + 4 + 4/3 + \dots$

Solution:

$$a = 12, r = 1/3 \quad S_\infty = 12/(1 - 1/3) = 12/(2/3) = 18$$

Example 4

R20,000 invested at 8% per year compounded quarterly for 5 years. Find final amount.

Solution:

$$A = 20000(1 + 0.08/4)^{(4 \times 5)} = 20000(1.02)^{20} = 20000 \times 1.4859 \approx R29,718.95$$

2.2 Practice

Independent Practice

1. Find the 10th term of: 7, 11, 15, 19, ...

Answer: _____

2. Find sum of first 20 terms of: 3, 7, 11, 15, ...

Answer: _____

3. Find T_n and S_n of: 2, 6, 18, 54, ...

Answer: _____

4. Does this series converge? $1 + 1/2 + 1/4 + 1/8 + \dots$ If yes, find sum.

Answer: _____

5. Find the sum to infinity: $100 - 50 + 25 - 12.5 + \dots$

Answer: _____

6. How many terms needed for sum > 1000 in: $1 + 2 + 4 + 8 + \dots$?

Answer: _____

7. R50,000 invested at 6% compounded monthly for 3 years. Final amount?

Answer: _____

8. What principal earns R10,000 in 4 years at 5% compounded annually?

Answer: _____

9. Which is better: 9% compounded monthly or 9.2% compounded annually?

Answer: _____

10. A sequence has $T_1 = 7$ and $T_5 = 19$. Find a and d (arithmetic).

Answer: _____

Chapter 3

Differential Calculus

Learning Objectives

- Understand the derivative as a rate of change and gradient of a tangent
- Differentiate using first principles
- Apply differentiation rules: power, product, chain, quotient
- Find maxima, minima, and points of inflection
- Apply calculus to optimisation and rate problems

Key Formulas

- First Principles: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
- Power Rule: $\frac{d}{dx} (x^n) = nx^{n-1}$
- Constant Rule: $\frac{d}{dx} (c) = 0$
- Sum/Difference: $\frac{d}{dx} (f \pm g) = f' \pm g'$
- Product Rule: $\frac{d}{dx} (fg) = f'g + fg'$
- Chain Rule: $\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$
- Quotient Rule: $\frac{d}{dx} \left(\frac{f}{g} \right) = \frac{f'g - fg'}{g^2}$
- Stationary points: $f'(x) = 0$
- Max: $f'(x) = 0$ and $f''(x) < 0$
- Min: $f'(x) = 0$ and $f''(x) > 0$
- Inflection: $f''(x) = 0$ (and sign changes)

3.1 Worked Examples

Example 1

Differentiate from first principles: $f(x) = x^2$

Solution:

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{(x^2 + 2xh + h^2 - x^2)}{h} = \lim_{h \rightarrow 0} \frac{(2xh + h^2)}{h} \\ = \lim_{h \rightarrow 0} (2x + h) = 2x$$

Example 2

Find dy/dx for: $y = 3x^3 - 5x^2 + 7x - 2$

Solution:

$$dy/dx = 12x^2 - 10x + 7$$

Example 3

Find stationary points of $f(x) = 2x^3 - 9x^2 + 12x - 4$ and classify them.

Solution:

$$f'(x) = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x-1)(x-2) \text{ Stationary at } x = 1 \text{ and } x = 2 \\ f''(x) = 12x - 18 \quad f''(1) = -6 < 0 \rightarrow \text{LOCAL MAX at } (1, 1) \quad f''(2) = 6 > 0 \rightarrow \text{LOCAL MIN at } (2, 0)$$

Example 4

A rectangular box (no lid) has square base of side x and height h . Volume = 32. Minimise surface area.

Solution:

$$V = x^2h = 32 \rightarrow h = 32/x^2 \quad SA = x^2 + 4xh = x^2 + 4x(32/x^2) = x^2 + 128/x \quad dSA/dx = 2x - \\ 128/x^2 = 0 \quad 2x^3 = 128 \rightarrow x^3 = 64 \rightarrow x = 4 \quad h = 32/16 = 2 \quad \text{Min SA: } x = 4, h = 2$$

Example 5

Differentiate: $y = (3x + 1)^3$

Solution:

$$\text{Chain rule: } dy/dx = 5(3x+1) \times 3 = 15(3x+1)$$

Common Mistakes

- Stationary point $\neq$ maximum/minimum — always use second derivative test to classify!
- $d/dx(3x^2) = 12x^3$, NOT $12x^2$ — bring the power DOWN and reduce by 1.

3.2 Practice

Independent Practice

1. Differentiate from first principles: $f(x) = 3x + 2$

Answer: _____

2. Find dy/dx : $y = 5x^3 - 4x^2 + 6x - 1$

Answer: _____

3. Find dy/dx : $y = \sqrt{x} + 1/x^2$

Answer: _____

4. Find all stationary points of: $f(x) = x^3 - 3x$ and classify.

Answer: _____

5. Differentiate using chain rule: $y = (2x^2 - 1)^3$

Answer: _____

6. Find the gradient of $y = x^3 - 4x$ at $x = 2$.

Answer: _____

7. A stone thrown up: $h = 20t - 5t^2$. Find max height and time to reach it.

Answer: _____

8. Find equation of tangent to $y = x^2 + 3x$ at $x = 1$.

Answer: _____

9. Find point of inflection of: $f(x) = x^3 - 6x^2 + 9x + 2$

Answer: _____

10. Differentiate: $y = (x+1)(x-3)$

Answer: _____

3.3 Exit Ticket

Exit Ticket

1. Differentiate: $y = 4x^3 - 2x$

Answer: _____

2. At what x is the gradient of $y = x^2 - 6x$ equal to zero?

Answer: _____

3. Is $(1, f(1))$ a max or min for $f(x) = -x^2 + 2x$? Why?

Answer: _____

4. Differentiate: $y = (x - 2)^3$ using chain rule.

Answer: _____

5. Write the derivative formula for the power rule.

Answer: _____

Chapter 4

Integral Calculus

Learning Objectives

- Understand integration as the reverse of differentiation (antiderivative)
- Apply indefinite and definite integration rules
- Calculate area under a curve using definite integrals
- Find area between two curves

Key Formulas

- Power Rule: $\int x^n dx = x^{n+1}/(n+1) + C$ ($n \neq -1$)
- $\int k dx = kx + C$
- $\int (f \pm g) dx = \int f dx \pm \int g dx$
- Definite integral: $\int [a \text{ to } b] f(x) dx = F(b) - F(a)$ where $F' = f$
- Area under curve: $A = \int [a \text{ to } b] |f(x)| dx$
- Area between curves: $A = \int [a \text{ to } b] (f(x) - g(x)) dx$ where $f(x) \geq g(x)$
- $\int e^x dx = e^x + C$ | $\int (1/x) dx = \ln|x| + C$

4.1 Worked Examples

Example 1

Find $\int (4x^3 - 6x + 5) dx$

Solution:

$$= 4x^4/4 - 6x^2/2 + 5x + C = x^4 - 3x^2 + 5x + C$$

Example 2

Evaluate $\int[1 \text{ to } 3] (2x + 1) dx$

Solution:

$$F(x) = x^2 + x \quad F(3) - F(1) = (9+3) - (1+1) = 12 - 2 = 10$$

Example 3

Find the area between $y = x^2$ and $y = x + 2$ from $x = -1$ to $x = 2$.

Solution:

Find intersections: $x^2 = x+2 \rightarrow x^2 - x - 2 = 0 \rightarrow (x-2)(x+1) = 0 \rightarrow x = -1, 2$ **Check:** $x+2 \geq x^2$ on $[-1, 2]$ $A = \int[-1 \text{ to } 2] (x+2 - x^2) dx = [x^2/2 + 2x - x^3/3] \text{ from } -1 \text{ to } 2 = (2 + 4 - 8/3) - (1/2 - 2 + 1/3) = (18/3) - (-7/6) = 6 + 7/6 = 43/6 \approx 7.17 \text{ sq units}$

Example 4

A particle moves with velocity $v(t) = 3t^2 - 4t$. Find displacement from $t = 0$ to $t = 3$.

Solution:

$$s = \int[0 \text{ to } 3] (3t^2 - 4t) dt = [t^3 - 2t^2] \text{ from } 0 \text{ to } 3 = (27 - 18) - 0 = 9 \text{ units}$$

4.2 Practice**Independent Practice**

1. Find $\int(6x^2 - 4x + 3) dx$

Answer: _____

2. Find $\int(\sqrt{x} + 1/x^2) dx$

Answer: _____

3. Evaluate $\int[0 \text{ to } 2] (x^2 + 1) dx$

Answer: _____

4. Evaluate $\int[1 \text{ to } 4] (2/x) dx$

Answer: _____

5. Find area under $y = x^2 + 1$ from $x = 0$ to $x = 3$.

Answer: _____

6. Find area between $y = x$ and $y = x^2$.

Answer: _____

7. If $f'(x) = 3x^2 - 2x$ and $f(0) = 5$, find $f(x)$.

Answer: _____

8. Evaluate $\int_{-1}^1 (x^3 + x) dx$ — explain the result geometrically.

Answer: _____

9. Find $\int (2x - 1)^3 dx$ (use substitution: $u = 2x - 1$)

Answer: _____

10. A car decelerates at $a(t) = -6t \text{ m/s}^2$. If $v(0) = 30$, when does it stop?

Answer: _____

Chapter 5

Statistics & Data Analysis

Learning Objectives

- Calculate measures of spread: variance and standard deviation
- Interpret and construct box-and-whisker plots
- Understand normal distribution and empirical rule
- Calculate and interpret linear regression and correlation coefficient

Key Formulas

- Mean: $\bar{x} = \Sigma x / n$
- Variance: $\sigma^2 = \Sigma (x - \bar{x})^2 / n$
- Standard Deviation: $\sigma = \sqrt{\Sigma (x - \bar{x})^2 / n}$
- Five-number summary: Min, Q1, Median (Q2), Q3, Max
- IQR = Q3 – Q1
- Outlier bounds: below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$
- Normal Distribution: 68% within 1σ , 95% within 2σ , 99.7% within 3σ
- Regression line: $\hat{y} = a + bx$ where b = correlation slope
- Correlation r : $-1 \leq r \leq 1$ (r near ± 1 = strong, r near 0 = weak)

5.1 Worked Examples

Example 1

Data: 4, 7, 7, 8, 10, 12, 15. Find variance and standard deviation.

Solution:

Mean = $63/7 = 9$ Deviations²: 25, 4, 4, 1, 1, 9, 36 Variance = $(25+4+4+1+1+9+36)/7 = 80/7 \approx 11.43$ SD = $\sqrt{11.43} \approx 3.38$

Example 2

Data: 3, 7, 8, 12, 15, 18, 20, 25. Draw box plot and identify outliers.

Solution:

Min=3, Q1=7.5, Median=13.5, Q3=19, Max=25 IQR = $19 - 7.5 = 11.5$ Lower bound: $7.5 - 1.75 \times 11.5 = -9.75$ Upper bound: $19 + 1.75 \times 11.5 = 36.25$ No outliers (all values within bounds)

Example 3

Test scores are normally distributed: mean 65, SD 8. What % of students scored between 49 and 81?

Solution:

$49 = 65 - 2(8) = \text{mean} - 2\sigma$ $81 = 65 + 2(8) = \text{mean} + 2\sigma$ By empirical rule: 95% of students scored between 49 and 81

5.2 Practice**Independent Practice**

1. Find mean, variance, and SD for: 5, 8, 10, 12, 15

Answer: _____

2. Draw a box plot for: 2, 5, 7, 8, 10, 12, 14, 18, 20

Answer: _____

3. Heights normally distributed: mean 170 cm, SD 6 cm. What % of people are between 158 cm and 182 cm?

Answer: _____

4. Find the IQR for: 3, 7, 9, 11, 14, 18, 22, 30

Answer: _____

5. Describe correlation for $r = -0.85$ and $r = 0.2$.

Answer: _____

6. Which data set has higher variability? A: SD = 2.1 B: SD = 5.8

Answer: _____

7. Test scores: mean 72, SD 10. What range contains 99.7% of scores?

Answer: _____

8. A data set has mean 50 and SD 5. A value of 35 is how many SDs from mean?

Answer: _____

9. Can standard deviation be negative? Explain.

Answer: _____

10. Describe what a regression line with slope 0 means.

Answer: _____

5.3 Exit Ticket

Exit Ticket

1. Data: 2, 4, 4, 6, 8. Find the mean and variance.

Answer: _____

2. What does a standard deviation of 0 tell you about a data set?

Answer: _____

3. In a normal distribution, what % of data falls within 1 SD of the mean?

Answer: _____

4. If $Q1 = 20$ and $Q3 = 40$, what is the IQR?

Answer: _____

5. Interpret: $r = 0.95$ between study hours and test scores.

Answer: _____

Answer Key — Selected Problems

Ch.1 #3: $\log_{10}(1000) - \log_{10}(10) = 3 - 1 = 2$

Ch.1 #4: $3^x = 81 = 3^4 \rightarrow x = 4$

Ch.1 #9: $e^{(2x)} = 7 \rightarrow 2x = \ln 7 \rightarrow x = \ln 7 / 2 \approx 0.973$

Ch.2 #1: $T = 7 + 9(4) = 43$

Ch.2 #2: $S = 20/2 \times (6 + 76) = 10 \times 82 = 820$

Ch.2 #4: $a=1, r=1/2 \rightarrow S_{\infty} = 1/(1-0.5) = 2$

Ch.3 #2: $dy/dx = 15x^2 - 8x + 6$

Ch.3 #4: $f'(x) = 3x^2 - 3 = 0 \rightarrow x = \pm 1$; max at $(-1, 2)$, min at $(1, -2)$

Ch.3 #7: $t = 2s$; max $h = 20$ m

Ch.4 #1: $x^3 - 2x^2 + 3x + C$

Ch.4 #3: $F(2) - F(0) = (8/3 + 2) - 0 = 14/3 \approx 4.67$

Ch.5 #1: Mean=10, Variance=13.2, SD $\approx$ 3.63

Ch.5 #3: $158 = 170 - 2(6)$ and $182 = 170 + 2(6) \rightarrow 95\%$

Ch.5 #8: $(35-50)/5 = -3$ SDs below the mean